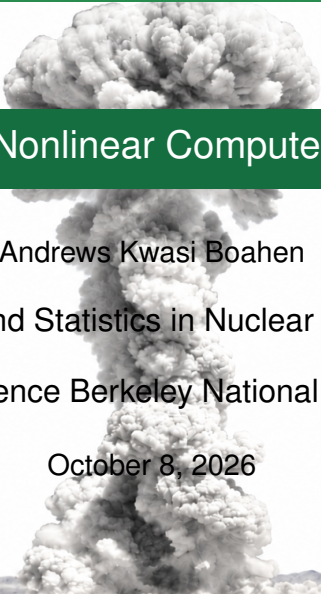




Active Learning for Nonlinear Computer Model Calibration

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ISNET-12: Information and Statistics in Nuclear Experiment and Theory

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Collaborators



Chih-Li Sung (Advisor).



Junoh Heo
(Assistant Professor, Wake Forest University)

What are Computer Experiments?

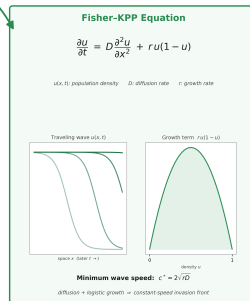
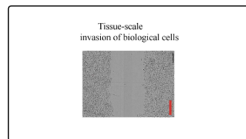
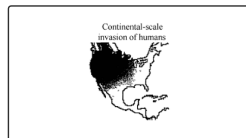
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Computer Experiments



What is Calibration?

- **Calibration** is variously referred to as model fitting, parameter estimation/inference, an inverse problem, and model tuning [1, An Introduction to the Calibration of Computer Models, 2023].

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- **Calibration** is variously referred to as model fitting, parameter estimation/inference, an inverse problem, and model tuning [1, An Introduction to the Calibration of Computer Models, 2023].
- It involves **adjusting or tuning** a set of parameters in a computer code, called calibration parameters, **using observations from physical experiments**, to make its outputs **emulate** reality.

Why does Computer Model Calibration Matter?

- Computer model calibration matters because simulators are **never quite exact!**

Why does Computer Model Calibration Matter?

- Computer model calibration matters because simulators are **never quite exact!**
- Knowing exactly **how wrong they are and where** is often more valuable than the simulation itself!

Simulators Aren't Truth

Every model is an approximation: **discretization, missing physics, unknown parameters.**

[2, "Science and statistics", 1976]

Well-calibrated models can be reused in different contexts [4, “Bayesian Calibration in a Multi-Output Transposition Context”, 2024].

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- In addition to $\mathcal{D}_n^p$, suppose we have a computer code f simulating the output of the physical phenomenon at locations $\{(x_i^*, t_i), x_i^* \in \Omega, t_i \in T\}_{i=1}^N$, where t_i is the setting of the calibration parameter θ for the i -th run.
- We denote the computer output by $z_i = f(x_i^*, t_i)$ and denote the simulation data by $\mathcal{D}_N^s = \{(x_i^*, t_i, z_i)\}_{i=1}^N$.

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- Goal: Integrating both data to improve prediction of η and reliability.

Statistical Emulation & Calibration

Gaussian Processes (GPs)

- GPs help encode prior information about functions.
- $f \sim \mathcal{GP}(\mu, K)$ with mean function μ and kernel K if $\forall$ finite set of inputs $\{x_1, \dots, x_n\}$, the vector

$$f(\mathbf{x}) = (f(x_1), \dots, f(x_n)) \sim \mathcal{N}(\boldsymbol{\mu}(\mathbf{x}), K(\mathbf{x}, \mathbf{x})),$$

- Prediction is straightforward using kriging equations:

$$\begin{aligned}\mu'(x^*) &= \mu(x^*) + K(x^*, X)K(X, X)^{-1}(Y - \mu(X)) \\ \sigma'^2(x^*) &= K(x^*, x^*) - K(x^*, X)K(X, X)^{-1}K(X, x^*)\end{aligned}$$

Gaussian Processes (GPs)

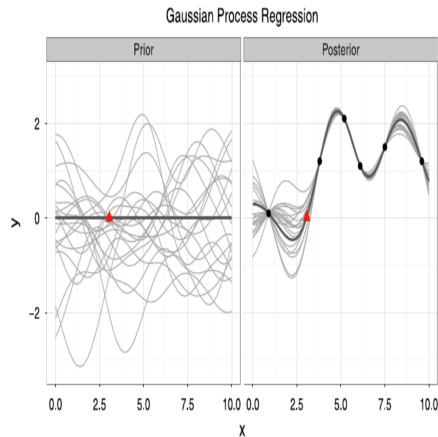
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Orthogonal Gaussian Processes

In universal kriging, $y(x) = \beta^\top g(x) + z(x)$, an ordinary GP prior lets $z(\cdot)$ spread into the trend subspace $M = \text{span}\{g(x)\}$, aliasing with β .

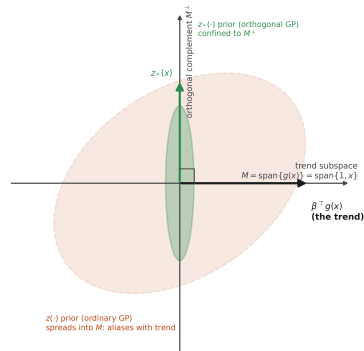
Fix: restrict $z_*(\cdot)$ to $M^\perp$,

$$\int_{\mathcal{X}} g(\xi) z_*(\xi) d\xi = 0 \text{ a.s.}$$

via

$$c_*(x, x') = c(x, x') - h(x)^\top H^{-1} h(x').$$

Variation resembling $g(x)$ must now come from β , not noise.



Deep Gaussian Processes

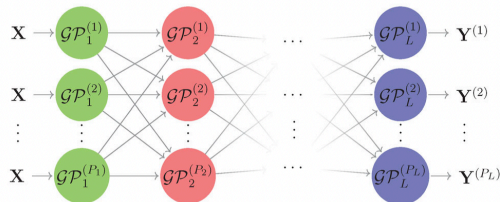
A deep GP is a composition of GP layers :

$$h^{(1)}(x) \sim \mathcal{GP}(0, K_1(x, x')),$$

$$h^{(\ell)}(x) \sim \mathcal{GP}(0, K_\ell(h^{(\ell-1)}(x), h^{(\ell-1)}(x'))), \quad \ell = 2, \dots,$$

$$y(x) = h^{(L)}(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

Each layer warps its input nonlinearly before the next GP acts on it. This lets the model capture non-stationary behavior a single GP layer cannot.



GP vs Deep GP [Example from Prof. Lawrence' s Lecture]

- Gold medal times for Olympic Marathon since 1896.

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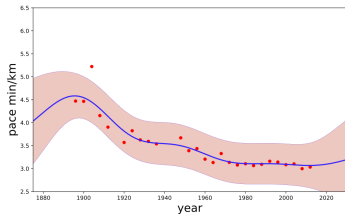


Figure: GP fit to the Olympic marathon data

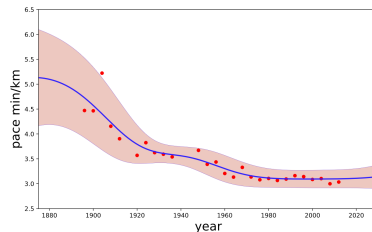


Figure: Deep GP fit to the Olympic marathon data

The Kennedy-O'Hagan Approach

- Additive Model:

$$y(\mathbf{x}) = f(\mathbf{x}, \boldsymbol{\theta}) + \delta(\mathbf{x}) + \epsilon(\mathbf{x}) \quad \mathbf{x} \in \Omega, \quad (1)$$

where δ is a discrepancy function and ϵ a Gaussian noise.

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- GP emulators.... Compute

$$\pi(\boldsymbol{\theta}|\mathbf{y}) \propto \frac{\pi(\boldsymbol{\theta})}{|\tau^2(\mathbf{K} + \mathbf{g}I)|^{1/2}} \exp \left\{ -\frac{1}{2\tau^2} (\mathbf{y} - \mathbf{f}_{\boldsymbol{\theta}})^T (\mathbf{K} + \mathbf{g}I)^{-1} (\mathbf{y} - \mathbf{f}_{\boldsymbol{\theta}}) \right\}$$
$$\pi(\delta(\mathbf{x}_0)|\mathbf{y}) \propto \int_{\Theta} \pi(\delta(\mathbf{x}_0)|\mathbf{y}, \boldsymbol{\theta}) \pi(\boldsymbol{\theta}|\mathbf{y}) d\boldsymbol{\theta},$$

- Markov Chain Monte Carlo, Modularization, ...

Other Approaches & Applications

- **Engineering** (Higdon et al. (2006); Gattiker et al. (2006; Higdon et al. (2013); Ling, Mullins, and Mahadevan (2014; Koermer et al. (2024)),
- **Climatology** (Sansó, Forest, and Zantedeschi (2008); Salter et al. (2019)),

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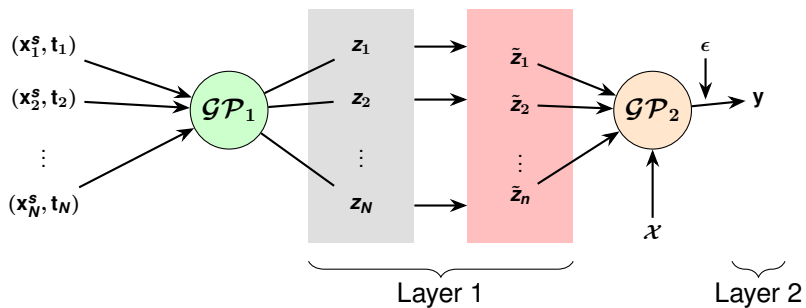
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- **Nuclear energy** [Higdon et al., 2015; Kejzlar et al., 2020,
- L_2 - calibration (Tuo and Wu), Orthogonal Gaussian Priors (Plumlee), Projected Kernel Calibration (Tuo)

Nonlinear Calibration

Model Architecture



Inference

- We use a "modularization" approach for the estimation procedure.

Inference

- Straightforward idea: To find $(\hat{\theta}, \hat{\zeta}_2)$, optimize:

$$(\hat{\theta}, \hat{\zeta}_2) = \operatorname{argmax}_{\theta, \zeta_2} \mathbb{E}_{\tilde{z}} [\boldsymbol{p}_{\theta, \zeta_2}(y|\tilde{z})] \quad (5)$$

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**Infinite mixture of Gaussians — Intractable —
Optimization challenging in high dimension**

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- We optimize a Monte Carlo version of the Evidence Lower Bound (ELBO):

$$(\hat{\theta}, \hat{\zeta}_2) = \operatorname{argmax}_{\theta, \zeta_2} \hat{\mathcal{L}}_{\text{ELBO}} \quad (7)$$

$$\hat{\mathcal{L}}_{\text{ELBO}} \propto -\frac{n}{2} \log(\tau_2^2) - \frac{1}{2M} \sum_{i=1}^M \log |\mathbf{L}_i| - \frac{1}{M} \sum_{i=1}^M \frac{\tau_2^{-2}}{2} (y - m_2 \mathbf{1}_n)^T \mathbf{H}_2, \quad (8)$$

- We wrap the hyperparameters' optimization in a routine R call via *optim* whereas we use **Simultaneous Perturbation Stochastic Approximation** for θ .

Training and Calibration: Monte-NAC with SPSA

Idea: maximize a Monte Carlo ELBO over θ , propagating layer-1 uncertainty through samples of $\tilde{z}$, using a gradient-free stochastic optimizer.

- 1 **Layer 1:** fit $\mathcal{GP}_1$ on $\mathcal{D}_N^s$ to obtain $\hat{\zeta}_1$.

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2 **Inner objective at θ :**

- Draw $\tilde{z}_i \sim \mathcal{N}(m_1^*, \Sigma_1^*)$, $i = 1, \dots, M$, from the $\mathcal{GP}_1$ posterior at (X^p, θ) .
- Closed-form $\hat{m}_2$, $\hat{\tau}_2^2$; numerically $\hat{\eta}_2 = \operatorname{argmax}_{\eta_2} \hat{\mathcal{L}}_{\text{ELBO}}$.
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3 **Outer SPSA on θ :** with Rademacher perturbation Δ_t and common random numbers,

$$\hat{g}_{t,j} = \frac{\hat{\mathcal{L}}(\theta_t + c_t \Delta_t) - \hat{\mathcal{L}}(\theta_t - c_t \Delta_t)}{2c_t \Delta_{t,j}}, \quad \theta_{t+1} = \Pi_{[l,u]}[\theta_t + a_t \hat{g}_t]$$

- Only **2** objective evaluations per iteration, regardless of $\dim(\theta)$.

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- Only 2 objective evaluations per iteration, regardless of $\dim(\theta)$.

Output: $\hat{\zeta}_1$, $\hat{\theta}$, $\hat{\zeta}_2$, and stored samples $\{\tilde{z}_i\}_{i=1}^M$ at $\hat{\theta}$ for prediction.

Prediction

- Idea: [Approximate equation \(6\) by a normal distribution](#) [20, [Deep Gaussian Process Emulation using Stochastic Imputation, 2022](#)],
- Assuming all hyperparameters and the calibration parameter has been estimated, the posterior predictive distribution of $\varphi(\mathbf{x}_s)$ at a new location $\mathbf{x}_s \in \Omega$ can be approximated, using M samples from the predictive distribution in layer 1 by

$$p(\eta(\mathbf{x}_s) | \mathcal{D}_N^s, \mathcal{D}_n^p, \mathbf{x}_s) \approx \frac{1}{M} \sum_{i=1}^M \mathcal{N}(m_2^*(\mathbf{x}_s), \sigma_2^{*2}(\mathbf{w}_s)), \quad (9)$$

where $m_2^*(\mathbf{x}_s)$ and $\sigma_2^{*2}(\mathbf{w}_s)$ are given by the following lemma.

Prediction

Lemma (posterior mean and variance using a sample path and SE kernel)

Let $\tilde{z}^{(i)} = (\tilde{z}_1, \dots, \tilde{z}_n)$ be a sample path from the predictive distribution of $\mathcal{GP}_1$ at $\hat{\theta}$, let $m_1^*(x_s, \hat{\theta})$ and $\sigma_1^{*2}(x_s, \hat{\theta})$ be the predictive mean and variance respectively from $\mathcal{GP}_1$ at $(x_s, \hat{\theta})$, let h_{2i} denote the i -th entry of h_2 and let $l_{ij}^{(-1)}$ be the (i, j) -th entry of the inverse of the $n \times n$ matrix $L_{\hat{\theta}}$. Under the squared exponential kernel, the posterior mean and variance of $y_s = \varphi(x_s)$ given $(\mathcal{D}_n^p)$ and simulation $(\mathcal{D}_N^s)$ data can be expressed respectively as:

$$\begin{aligned} \mu_2^*(x_s) &= \hat{\mu}_2 + \sum_{i=1}^n h_{2i} \prod_{j=1}^d \exp\left(-\frac{(x_{sj} - x_{ij})^2}{\hat{\zeta}_{2,j}}\right) \\ &\quad \times \sqrt{\frac{\hat{\eta}_{2,j+1}}{\hat{\zeta}_{2,j+1} + 2\sigma_1^{*2}(x_s, \hat{\theta})}} \exp\left(-\frac{(\tilde{z}_i - \mu_1^*(x_s, \hat{\theta}))^2}{\hat{\zeta}_{2,j+1} + 2\sigma_1^{*2}(x_s, \hat{\theta})}\right) \end{aligned} \quad (10)$$

and

$$\begin{aligned} \sigma_2^{*2}(x_s) &= \hat{\tau}_2^2 - (\mu_2^*(x_s) - \mu_2)^2 + \left[\sum_{i,k=1}^n \nu_{ik} (h_{2i} h_{2k} - \hat{\tau}_2^2 (L_2(\hat{\theta})^{-1})_{ik}) \right. \\ &\quad \left. \times \prod_{j=1}^d \exp\left(-\frac{(x_{sj} - x_{ij})^2 + (x_{sj} - x_{kj})^2}{\hat{\zeta}_{2,j}}\right) \right], \end{aligned} \quad (11)$$

Prediction

where,

$$\nu_{ik} = \sqrt{\frac{\hat{\zeta}_{2,j+1}}{\hat{\zeta}_{2,j+1} + 4\sigma_1^{*2}(\mathbf{x}_s, \hat{\theta})}} \exp\left(-\frac{\left(\frac{\tilde{z}_i + \tilde{z}_k}{2} - \mu_1^*(\mathbf{x}_s, \hat{\theta})\right)^2}{\frac{\hat{\eta}_{2,j+1}}{2} + 2\sigma_1^{*2}(\mathbf{x}_s, \hat{\theta})} - \frac{(\tilde{z}_i - \tilde{z}_k)^2}{2\hat{\zeta}_{2,j+1}}\right). \quad (12)$$

- The proof of this lemma is based on expanding the variance of the prediction y_s as

$$\mathbb{V}[y_s | \mathcal{D}_n^p, \mathcal{D}_N^s, \mathbf{x}_s] = \underbrace{\mathbb{V}[E[y_s | \mathcal{D}_n^p, \mathcal{D}_N^s, \mathbf{x}_s, \mathbf{z}_s]]}_{\text{contribution by } \mathcal{GP}_1} + \underbrace{\mathbb{E}[V[y_s | \mathcal{D}_n^p, \mathcal{D}_N^s, \mathbf{x}_s, \mathbf{z}_s]]}_{\text{contribution by } \mathcal{GP}_2}$$

- This decomposition of the variance is similar to **Ming and Guillas** (2021) within the context of linked GPs; **Heo and Sung** (2025) for the RNA emulator.

Prediction

Lemma (posterior mean and variance of the NAC emulator using SE Kernel)

Under the conditions of the previous Lemma, suppose we have sample paths $\tilde{z}_i, i = 1, \dots, M$. The posterior mean and variance of the NAC emulator are given by :

$$\mu_2^*(\mathbf{x}_s) = \frac{1}{M} \sum_{i=1}^M \mu_2^{*(i)}(\mathbf{x}_s) \quad \text{and}$$

$$\sigma_2^{*2}(\mathbf{x}_s) = \frac{1}{M} \sum_{i=1}^M \left(\mu_2^{*(i)}(\mathbf{x}_s)^2 + \sigma_2^{*(i)}(\mathbf{x}_s)^2 \right) - \mu_2^*(\mathbf{x}_s)^2,$$

where $\mu_2^{(i)}(\mathbf{x}_s)$ and $\sigma_2^{*(i)}(\mathbf{x}_s)^2$ are the posterior mean and variance in the previous Lemma using $\tilde{z}_i$*

OGP Kernel Construction

- Recall that the main goal of calibration is to find θ^* :

$$\theta^* = \operatorname{argmin}_{t \in \Theta} \int_{\mathcal{X}} (\eta(\xi) - f(\xi, t))^2 d\mu(\xi) \quad (13)$$

where μ is a weighing scheme on $\mathcal{X}$.

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where μ is a weighing scheme on $\mathcal{X}$.

- Assume f is computationally "cheap", and $f(\mathbf{x}, t)$ is differentiable in t at θ^* for all $\mathbf{x} \in \Omega$. Under a set of assumptions, Eq(13) admits a unique solution if

$$\int (g(\mathbf{x}, f(\mathbf{x}, \theta)) - f(\mathbf{x}, \theta^*)) Df(\mathbf{x}, \theta^*) d\mathbf{x} = 0. \quad (14)$$

OGP Kernel Construction

- let $z_\theta(\xi) = f(\xi, \theta)$ and define the functions

$$h_\theta(w) = h_\theta((\xi, z_\theta(\xi))) = \int_{\mathcal{X}} K_2((\xi, z_\theta(\xi)), (x, z_\theta(x))) \nabla z_\theta(\xi) d\xi,$$

and

$$H_\theta = \int_{\mathcal{X}} \int_{\mathcal{X}} K_2((\xi, z_\theta(\xi)), (\xi', z_\theta(\xi'))) \nabla z_\theta(\xi) \nabla z_\theta(\xi')^T d\xi d\xi',$$

we choose the kernel of g to be

$$K_\theta(w, w') = K_2(w, w') - h_\theta^T(w) H_\theta^{-1} h_\theta(w'). \quad (15)$$

- We derive the following theorem

OGP Kernel Construction

Theorem

Suppose $\mathcal{X}$ is a bounded subset of $\mathbb{R}^d$, $Df(\cdot)$ is bounded on $\mathcal{X}$ and $K(\cdot, \cdot)$ is bounded and continuous on $(\mathcal{X}, \mathbb{R}) \times (\mathcal{X}, \mathbb{R})$, then:

- i) $K_\theta(w, w')$ in Equation (15) is a semi-positive definite function on $(\mathcal{X}, \mathbb{R}) \times (\mathcal{X}, \mathbb{R})$.
- ii) Additionally, if $g(x, f(x, \theta)) \sim \mathcal{GP}(f(x, \theta^*), K_\theta(w, w'))$, then,

$$P\left(\int (g(x, f(x, \theta)) - f(x, \theta^*)) Df(x, \theta^*) dx = 0\right) = 1 \quad (16)$$

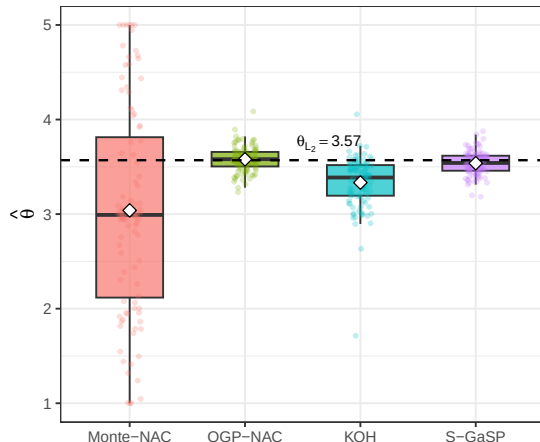
Numerical Example

- Consider $\eta(x) = 4x + x \sin(5x)$ and $f(x, t) = tx$ with $\mathcal{X} \times \mathcal{T} = [0, 1] \times \mathbb{R}$
- For our illustrative example, we compute the minimizer of the L^2 loss as

$$\theta_{L^2}^* = \operatorname{argmin}_{t \in \mathcal{T}} \int_0^1 (tx - 4x - x \sin(5x))^2 dx = \frac{\langle x, 4x + x \sin(5x) \rangle}{\langle x, x \rangle} \approx 3.57.$$

- We run 100 replications with $(n = 9, N = 15, \sigma = .2)$ and compare calibration and predictive performances of our methods against KOH and S-GaSP.

Numerical Example (Calibration Performance)

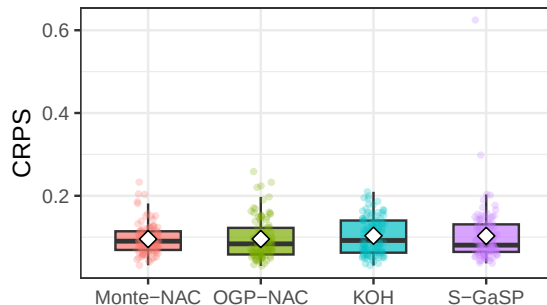
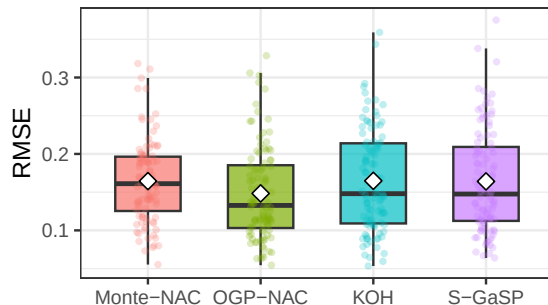


- Across 100 replications, OGP-NAC has the smallest estimation bias.
- OGP-NAC and S-GaSP estimates are closer to θ_{L_2} as expected.
- Diamonds represent the mean value

Method	Mean	SD	Bias	RMSE
Monte-NAC	3.040	1.090	-0.531	1.210
OGP-NAC	3.580	0.135	0.010	0.135
KOH	3.330	0.281	-0.237	0.366
S-GaSP	3.540	0.128	-0.032	0.131

$$\theta_{L_2} = 3.57, \quad R = 100.$$

Numerical Example (Predictive Performance)



OGP-NAC and Monte-NAC have slightly better predictive performance than KOH and Scaled-KOH.

Takeaway 1

“ Our NAC methods provide competitive predictive accuracy and improved probabilistic prediction. The orthogonal kernel construction additionally resolves calibration non-identifiability, recovering θ_{L_2} without sacrificing predictive performance.”

Active Learning

Active Learning

...The best time to design an experiment is when it is done, the converse of which is the worst time to design an experiment is at the beginning. (Box et al, 1978)

- The objective is to determine the next optimal point, either $\mathbf{x}_{n+1} \in \Omega$ or $(\mathbf{x}_{N+1}, t_{N+1}) \in \Omega \times \mathbb{R}$, that should be added to the initial designs $\mathcal{D}_N^s, \mathcal{D}_n^p$ to improve the performance of our emulator.

Active Learning

...The best time to design an experiment is when it is done, the converse of which is the worst time to design an experiment is at the beginning. (Box et al, 1978)

- The objective is to determine the next optimal point, either $\mathbf{x}_{n+1} \in \Omega$ or $(\mathbf{x}_{N+1}, t_{N+1}) \in \Omega \times \mathbb{R}$, that should be added to the initial designs $\mathcal{D}_N^s, \mathcal{D}_n^p$ to improve the performance of our emulator.
- Let C_1 and C_2 be the costs associated with collecting simulation data and physical data respectively.

Active Learning MacKay (ALM) for NAC

Goal: target the input with the largest predictive variance, then decide whether a physical or a simulation run reduces that variance more per unit cost.

At iteration t :

- ① **Physical candidate:** select field input

$$\mathbf{x}_{n+1} = \underset{\mathbf{x} \in \Omega}{\operatorname{argmax}} \sigma^{2*}(\mathbf{x}),$$

with variance reduction $\mathcal{B}_1(\mathbf{x}_{n+1}) = \sigma^{2*}(\mathbf{x}_{n+1}) - \sigma^{2*}(\mathbf{x}_{n+1} \mid \mathcal{D}_n^p \cup \{\mathbf{x}_{n+1}\}) .$

Active Learning MacKay (ALM) for NAC

Goal: target the input with the largest predictive variance, then decide whether a physical or a simulation run reduces that variance more per unit cost.

At iteration t :

- ① **Physical candidate:** select field input

$$x_{n+1} = \operatorname{argmax}_{x \in \Omega} \sigma^{2*}(x),$$

with variance reduction $\mathcal{B}_1(x_{n+1}) = \sigma^{2*}(x_{n+1}) - \sigma^{2*}(x_{n+1} \mid \mathcal{D}_n^p \cup \{x_{n+1}\})$.

- ② **Simulation candidate:** the run that best reduces variance at x_{n+1} ,

$$(x_{N+1}, t_{N+1}) = \operatorname{argmax}_{(x,t) \in \Omega \times T} \mathcal{B}_2(x, t),$$

where $\mathcal{B}_2(x, t) = \sigma^{2*}(x_{n+1}) - \sigma^{2*}(x_{n+1} \mid \mathcal{D}_N^s \cup \{(x, t)\})$.

Active Learning MacKay (ALM) for NAC

Goal: target the input with the largest predictive variance, then decide whether a physical or a simulation run reduces that variance more per unit cost.

At iteration t :

- ① **Physical candidate:** select field input

$$x_{n+1} = \operatorname{argmax}_{x \in \Omega} \sigma^{2*}(x),$$

with variance reduction $\mathcal{B}_1(x_{n+1}) = \sigma^{2*}(x_{n+1}) - \sigma^{2*}(x_{n+1} \mid \mathcal{D}_n^p \cup \{x_{n+1}\})$.

- ② **Simulation candidate:** the run that best reduces variance at x_{n+1} ,

$$(x_{N+1}, t_{N+1}) = \operatorname{argmax}_{(x,t) \in \Omega \times T} \mathcal{B}_2(x, t),$$

where $\mathcal{B}_2(x, t) = \sigma^{2*}(x_{n+1}) - \sigma^{2*}(x_{n+1} \mid \mathcal{D}_N^s \cup \{(x, t)\})$.

- ③ **Selection:**

$$C_2 \mathcal{B}_1(x_{n+1}) > C_1 \mathcal{B}_2(x_{N+1}, t_{N+1}) \Rightarrow \mathcal{D}_{n+1}^p \leftarrow \mathcal{D}_n^p \cup \{(x_{n+1}, y_{n+1})\},$$

otherwise

$$\mathcal{D}_{N+1}^s \leftarrow \mathcal{D}_N^s \cup \{(x_{N+1}, t_{N+1}, f(x_{N+1}, t_{N+1}))\}.$$

Active Learning Cohn (ALC)

Goal: At each step, add the point that most reduces the integrated predictive variance of the NAC emulator, and decide between running the simulator or collect a physical observation.

At iteration t :

① Fit $\mathcal{GP}_1$ and $\mathcal{GP}_2$ on the current data $\mathcal{D}_{N+t}^s$ and $\mathcal{D}_{n+t}^p$.

② **Simulation candidate:**

$$(\mathbf{x}_s^*, \boldsymbol{\theta}_s^*) = \arg \min_{(\mathbf{x}, \boldsymbol{\theta}) \in \Omega \times \Theta} \int_{\Omega} \sigma_2^{*2}(\mathbf{x}' \mid \mathcal{D}^p, \mathcal{D}^s \cup \{(\mathbf{x}, \boldsymbol{\theta})\}) d\mathbf{x}'$$

③ **Physical candidate:**

$$\mathbf{x}_p^* = \arg \min_{\mathbf{x} \in \Omega} \int_{\Omega} \sigma_2^{*2}(\mathbf{x}' \mid \mathcal{D}^p \cup \{\mathbf{x}\}, \mathcal{D}^s) d\mathbf{x}'$$

④ **Selection:** with variance reductions in step 2 (simulation) and step 3 (physical),

$$\frac{V_1}{C^1} > \frac{V_2}{C^2} \Rightarrow \text{run simulator at } (\mathbf{x}_s^*, \boldsymbol{\theta}_s^*), \quad \text{otherwise observe reality at } \mathbf{x}_p^*.$$

ALM experimental Setup

- Initial sample is generated via LHS with $N = 8, n = 5$.
- We have a budget of 20 new acquisitions with $C_1 = 1$ and $C_2 = 4$.
- OGP-NAC is used for inference
- Compare ALM and ALC acquisition strategies for Plumlee's function.

Figure 1 displays a comparison of the proposed algorithm with the baseline algorithm across three iterations: Iteration 1, Iteration 2, and Iteration 15. The figure is organized into two rows of plots.

The top row shows the **Response** (Y-axis) versus **x** (X-axis, ranging from 0.00 to 1.00). The plots compare the **Physical** (black line) and **Simulator** (blue line) data. The **Physical** data is fitted with a model $\hat{\theta}$, and the **Simulator** data is fitted with a model $\hat{\theta}$. The **RMSE** (Root Mean Square Error) is provided for each fit.

- Iteration 1:** Physical | $\hat{\theta} = 3.936$ | RMSE = 0.188
- Iteration 2:** Simulator | $\hat{\theta} = 3.862$ | RMSE = 0.204
- Iteration 15:** Physical | $\hat{\theta} = 3.463$ | RMSE = 0.090

The bottom row shows the **Acquisition score $v(x)$** (Y-axis) versus **x** (X-axis, ranging from 0.00 to 1.00). The plots compare the **Physical** (black line) and **Simulator** (blue line) data. The **Acquisition score** is provided for each fit.

- Iteration 1:** Physical: 4.27e-03 Simulator: 7.42e-04
- Iteration 2:** Physical: 9.59e-04 Simulator: 3.07e-03
- Iteration 15:** Physical: 1.92e-04 Simulator: 1.36e-05

The plots illustrate the performance of the proposed algorithm (Physical) compared to the baseline algorithm (Simulator) across different iterations. The **Acquisition score** is a key metric for evaluating the performance of the algorithm, and the plots show that the proposed algorithm (Physical) consistently achieves a higher acquisition score than the baseline algorithm (Simulator) across all iterations.

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Iteration 1
Physical | $\hat{\theta} = 3.981$ | RMSE = 0.191

Iteration 2
Simulator | $\hat{\theta} = 3.941$ | RMSE = 0.264

Iteration 18
Physical | $\hat{\theta} = 3.632$ | RMSE = 0.066

Select: Physical
Physical: $8.11\text{e-}04$ Simulator: $1.94\text{e-}04$

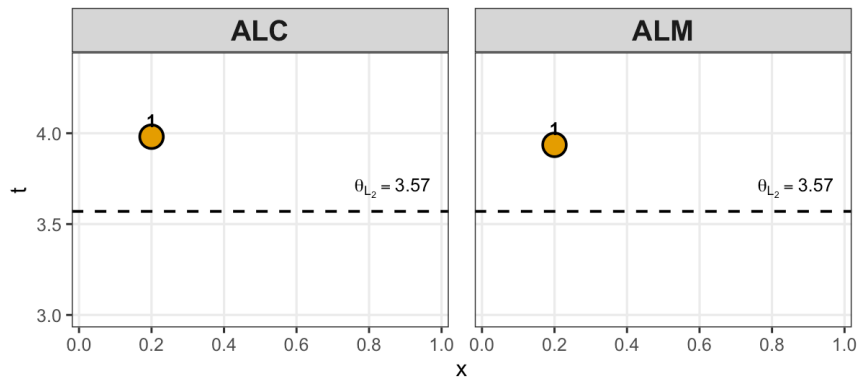
Select: Simulator
Physical: $3.04\text{e-}04$ Simulator: $1.13\text{e-}03$

Select: Physical
Physical: $4.74\text{e-}05$ Simulator: $5.83\text{e-}06$

Sequential Acquisition: ALM vs ALC

Acquisition 1

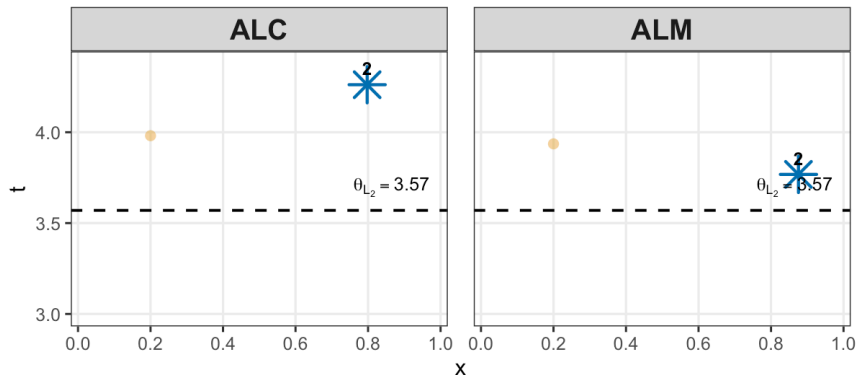
Physical acquisition: (x, $\hat{\theta}$) | Simulator acquisition: (x, t)



Sequential Acquisition: ALM vs ALC

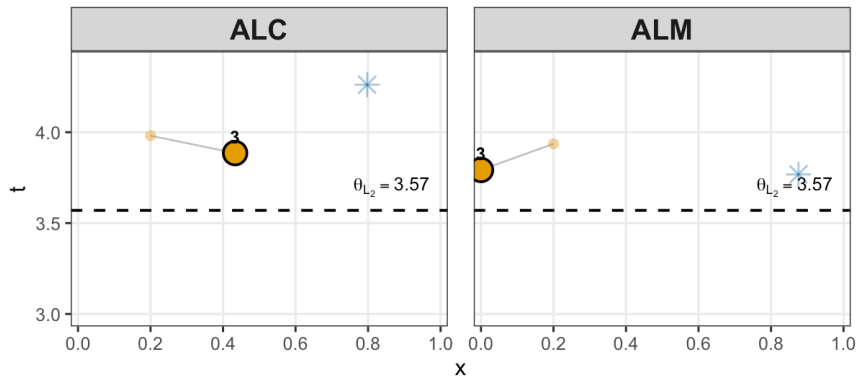
Acquisition 2

Physical acquisition: (x, theta-hat) | Simulator acquisition: (x, t)



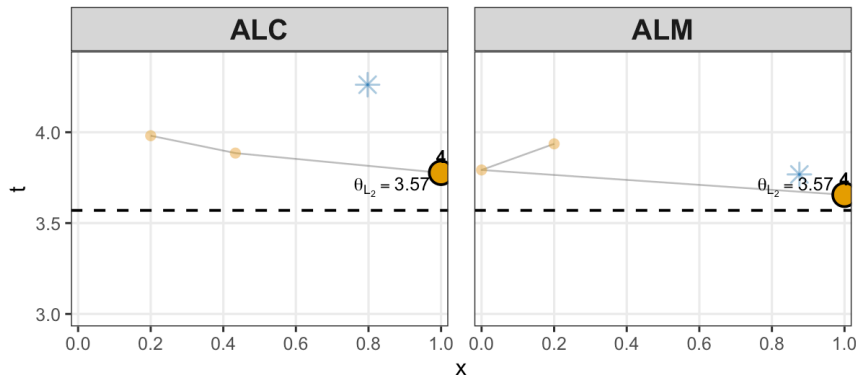
Sequential Acquisition: ALM vs ALC

Acquisition 3

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

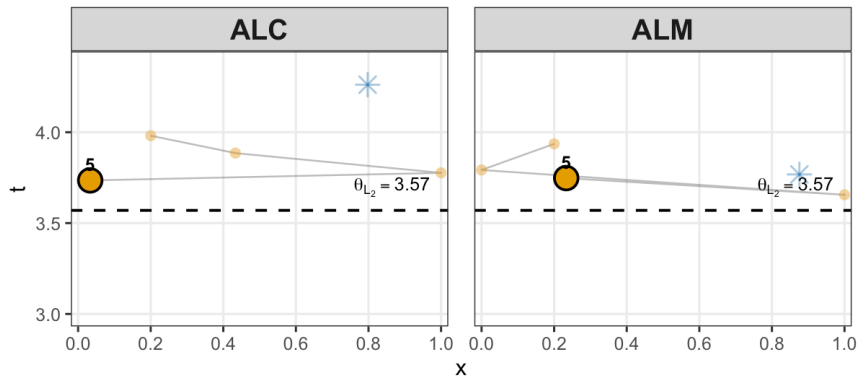
Sequential Acquisition: ALM vs ALC

Acquisition 4

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

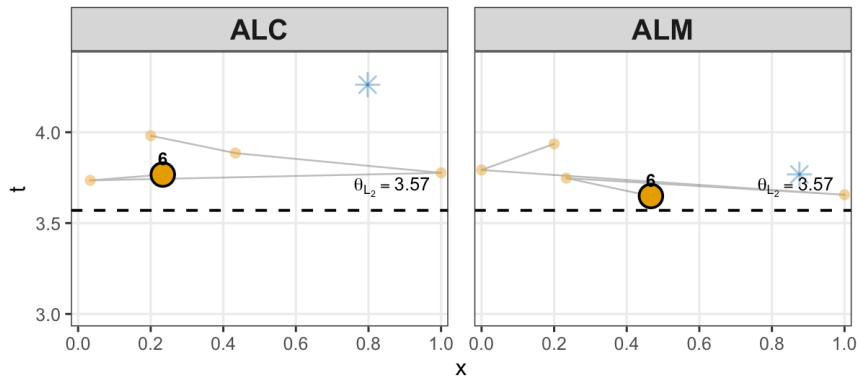
Acquisition 5

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

Acquisition 6

Physical acquisition: (x, θ -hat) | Simulator acquisition: (x, t)

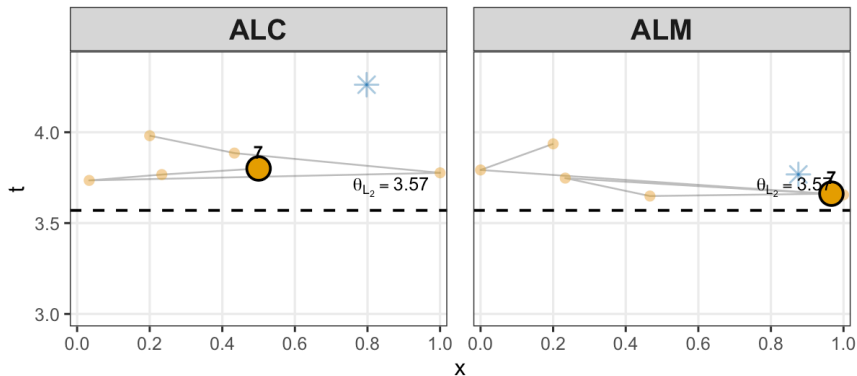


Sequential Acquisition: ALM vs ALC

Acquisition 7

Physical acquisition: (x, θ)

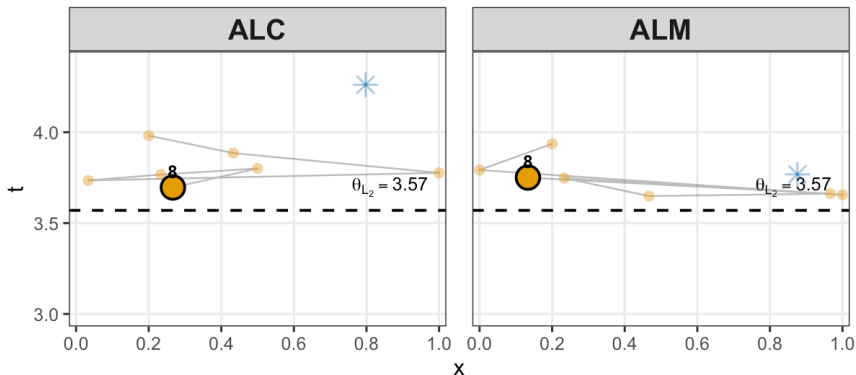
Simulator acquisition: (x, t)



Sequential Acquisition: ALM vs ALC

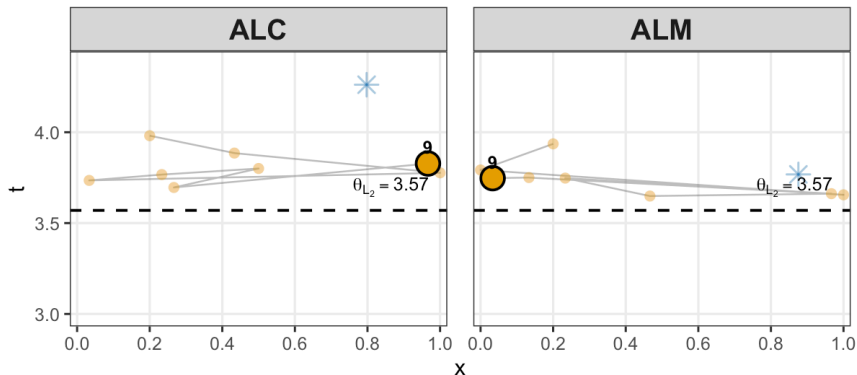
Acquisition 8

Physical acquisition: (x, $\hat{\theta}$) | Simulator acquisition: (x, t)



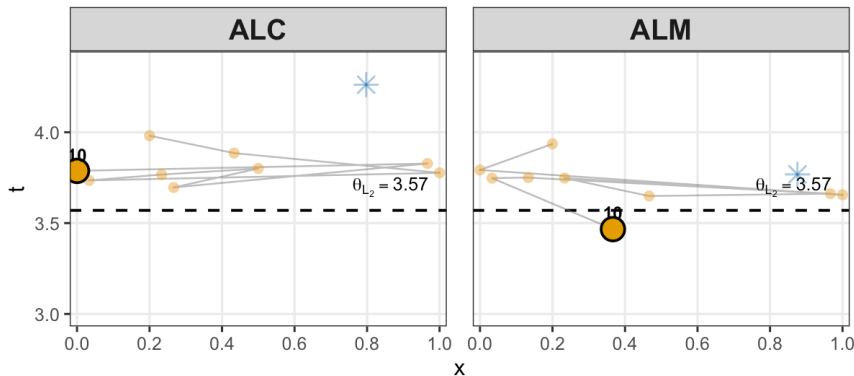
Sequential Acquisition: ALM vs ALC

Acquisition 9

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

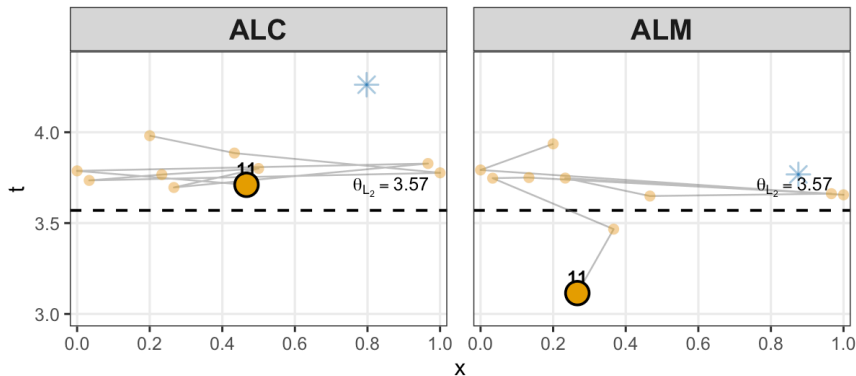
Acquisition 10

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

Acquisition 11

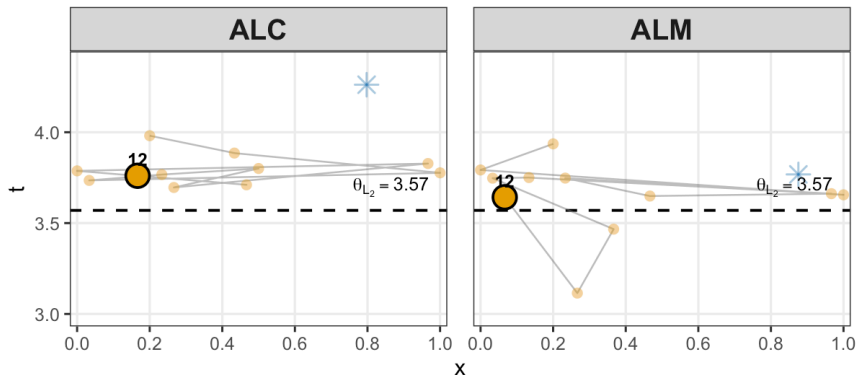
Physical acquisition: (x, theta-hat) | Simulator acquisition: (x, t)



Sequential Acquisition: ALM vs ALC

Acquisition 12

Physical acquisition: (x, θ -hat) | Simulator acquisition: (x, t)

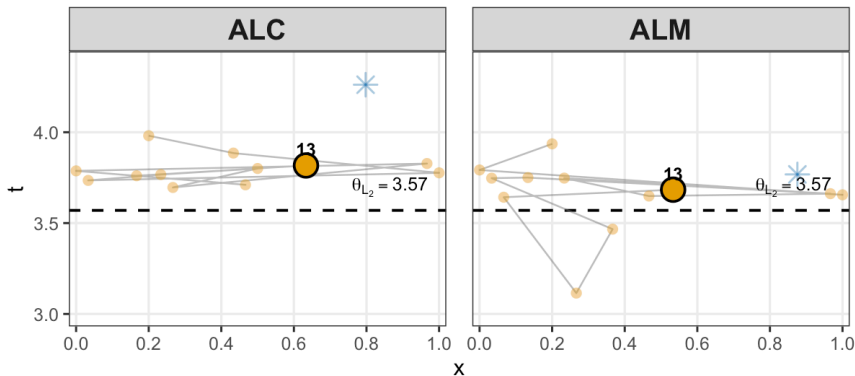


Sequential Acquisition: ALM vs ALC

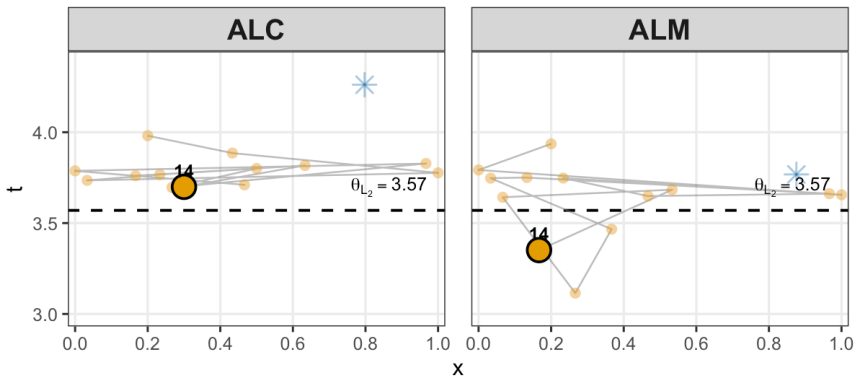
Acquisition 13

Physical acquisition: (x, theta-hat)

Simulator acquisition: (x, t)



Acquisition 14

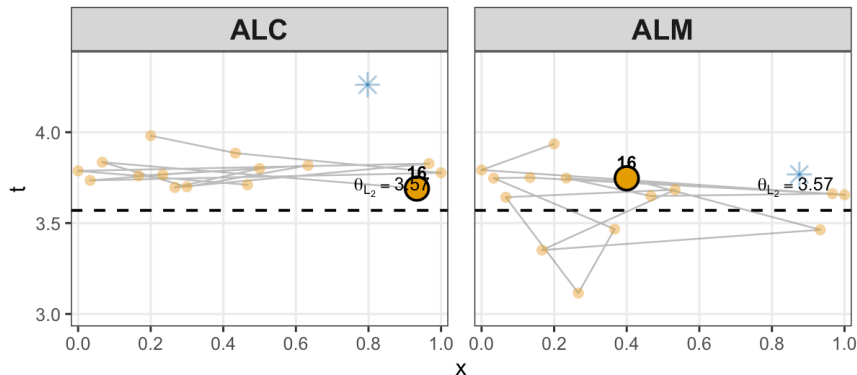
Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Acquisition 15

Sequential Acquisition: ALM vs ALC

Acquisition 16

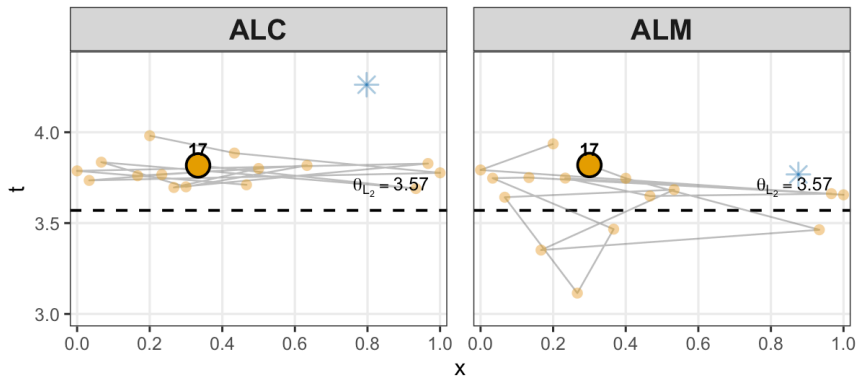
Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t)



Sequential Acquisition: ALM vs ALC

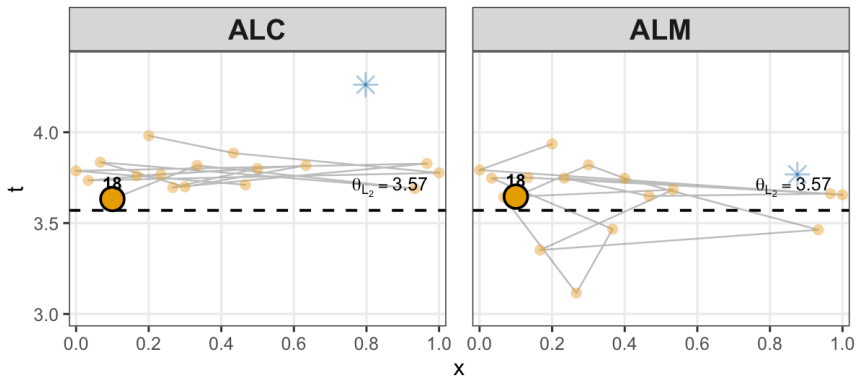
Acquisition 17

Physical acquisition: (x, θ -hat) | Simulator acquisition: (x, t)



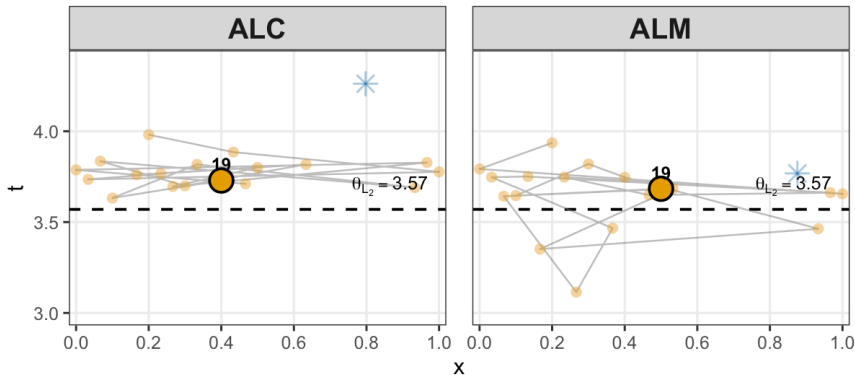
Sequential Acquisition: ALM vs ALC

Acquisition 18

Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

Acquisition 19

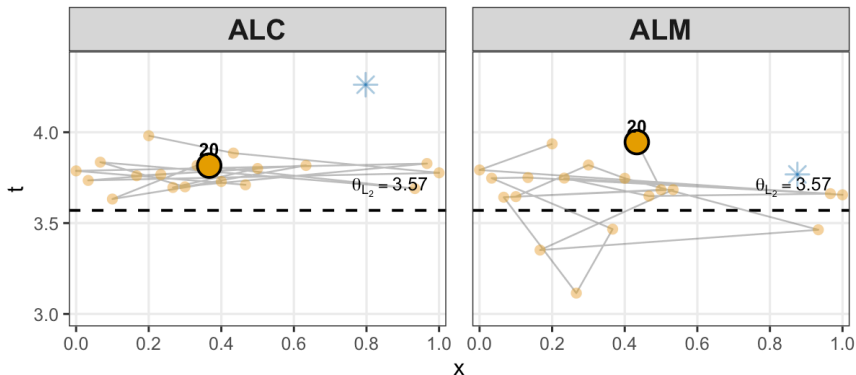
Physical acquisition: $(x, \hat{\theta})$ | Simulator acquisition: (x, t) 

Sequential Acquisition: ALM vs ALC

Acquisition 20

Physical acquisition: $(x, \hat{\theta})$

Simulator acquisition: (x, t)



Key Takeaways

- We introduce the NAC emulator for a generalized version of the KOH model together with active learning strategies to improve its predictive performance.
- Closed forms of the predictive mean and variance of the NAC emulator are useful not only for uncertainty quantification but can make active learning efficient
- We extend the Orthogonal GP method idea to the NAC framework.
- Currently, more physical data is required to improve the performance of the emulator.

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Thank you!

Questions are welcome

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